

Review

Regulation of mRNA stability in mammalian cells

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Abstract

The regulation of mRNA decay is a major control point in gene expression. The stability of a particular mRNA is controlled by specific interactions between its structural elements and RNA binding proteins that can be general or mRNA specific. Regulated mRNA stability is achieved through fluctuations in half lives in response to developmental or environmental stimuli like nutrient levels, cytokines, hormones and temperature shifts as well as environmental stresses like hypoxia, hypocalcemia, viral infection, and tissue injury. Furthermore, in specific disorders like some forms of neoplasia, thalassemia and Alzheimer's disease, deregulated mRNA stability can lead to the aberrant accumulation of mRNAs and the proteins they encode. This review presents a discussion of some recently identified examples of regulated and deregulated mRNA stability in order to illustrate the diversity of genes regulated by alterations in the degradation rates of their mRNAs. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: mRNA decay; Decay pathways; 3' Untranslated region; RNA binding proteins; Hormonal regulation of mRNA turnover; Deregulated decay

1. Introduction

For the myriad genes expressed within a single cell, the precise control of gene expression can present a formidable biochemical problem. In eukaryotes, the processes of transcription and translation are compartmentalized. This, together with the complexity of nuclear processing events required to generate a translatable mRNA, permits control to be accomplished at multiple points during the flow of genetic information from DNA to protein. In many cases, the regulation of protein production is principally achieved by controlling mRNA levels. Many previous studies in this

regard have focused on transcription and pre-mRNA processing. However, cytoplasmic mRNA levels represent a balance between the rates of nuclear RNA synthesis, processing, and export, and the rates of cytoplasmic mRNA degradation. Based on the ever-increasing volume of information that has been accumulating over the past two decades, the process of mRNA decay is now considered a major control point in gene expression.

In this review, we first describe how mRNA decay contributes to the level and timing of gene expression and describe the pathways by which mRNAs are degraded. The rates at which degradation processes can occur are dictated by both cis-acting elements within an mRNA and trans-acting factors that bind them. We will then provide a brief overview of some of these cis and trans components, our aim being to impart a flavor for the complexity and diversity of these components. Finally, we provide a discussion of selected examples of regulated mRNA stability. We hope to impress upon the reader an appreciation for the diversity of genes and biological systems that utilize mRNA turnover as a mechanism for controlling gene expression.

2. Significance of mRNA stability

The rate of mRNA turnover not only determines the rate of disappearance of mRNA but also its induction. mRNAs

Abbreviations: α CP, α globin specific stabilizing protein; α 1 RBF, α 1 procollagen RNA binding factor; ARE, A + U rich element; bp, base pair; COX, cyclooxygenase; CRE, C rich element; CYP7A1, cholesterol 7 α hydroxylase; ELAV, embryonic lethal abnormal vision; GAPDH, glyceraldehyde 3 phosphate dehydrogenase; GM-CSF, granulocyte monocyte colony stimulating factor; IFN, interferon; IL, interleukin; IRE, iron responsive element; IRP, iron regulatory protein; JNK, janus kinase; JRE, JNK response element; LDH, lactate dehydrogenase; NMD, nonsense mediated decay; nt, nucleotide; poly(A), polyadenylate; PABP, poly(A) binding protein; PARN, poly(A) ribonuclease; PCSR, PKC stabilizing region; PG, prostaglandin; PKA, protein kinase A; PKC, protein kinase C; PTH, parathyroid hormone; T₃, thyrotropin; TfR, transferrin receptor; TGF, transforming growth factor; TNF, tumor necrosis factor; TPA, 12 O tetradecanoylphorbol 13 acetate; TTP, tristetraprolin; UTR, untranslated region; VEGF, vascular endothelial growth factor

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with short half-lives respond to changes in transcription more rapidly than those that are relatively stable. (For further discussion, refer to Ross, 1995.) It is the wide range of mRNA decay rates in higher eukaryotes that contributes significantly to the regulation of gene expression. Differential mRNA turnover may contribute to rapid changes in the pattern of cellular gene expression in response to changing environmental or developmental cues on one hand and can rigidly maintain the levels of a translatable transcript on the other. Therefore, stable mRNAs permit a prolonged translational window for genes that are expressed at high levels, such as β -globin (Russell et al., 1997) as well as in cases when translation is delayed, such as maternal mRNAs, which are stably stored until after oocyte fertilization (Spirin, 1994). By contrast, mRNAs that encode proteins produced only in short bursts in response to internal or external stimuli have short half-lives (Schiavi et al., 1992; Ross 1995; Paulding and Czyzyk-Krzeska, 2000; Staton et al., 2000). These stimuli range from developmental (proliferating and differentiating), nutritional, hormonal, and pharmacological, to environmental alterations, such as temperature shifts and insults like hypoxia and viral infection. Examples of such unstable transcripts include mRNAs encoding cell-cycle regulated histones, oncoproteins, cytokines, and lymphokines.

3. mRNA decay pathways in mammalian cells

Several decay pathways have been characterized for mRNAs (Fig. 1). The purpose of this section is not to review in depth the recent findings in this area, but to briefly describe our present knowledge, prior to embarking upon a discussion of specific examples of regulated mRNA stability. Interested readers are, however, referred to some excellent reviews on the subject (Decker and Parker, 1994; Beelman and Parker, 1995; Ross, 1995; Liebhaver, 1997; Mitchell and Tollervy, 2000). Different mRNAs can be degraded via distinct pathways while a given mRNA can also be degraded by different, seemingly-redundant pathways, depending upon cellular conditions.

The regulated decay of mRNA is achieved by the orchestrated interactions between an mRNA's structural components and specific trans-acting factors. These components include the 5'-cap structure, 5'-untranslated region (UTR), the protein coding region, 3'-UTR and the 3'-polyadenylate [poly(A)] tail.

Most mammalian mRNAs are polyadenylated. The 3' poly(A) tail of eukaryotic mRNAs plays important roles in nuclear mRNA processing, export to the cytoplasm, translation and cytoplasmic mRNA stability (reviewed in Bernstein and Ross, 1989; Sachs, 1990). Probably most of the poly(A) tracts in mammalian mRNAs remain bound to the highly expressed poly(A)-binding protein (PABP), which exhibits a high affinity for poly(A) (Gorlach et al., 1994).

Based on various studies, the two observations that deadenylation is the first step in the decay of many mammalian mRNAs and that the poly(A)-PABP interaction at the 3' terminus protects mRNAs from rapid decay in vitro, have led to the conclusion that poly(A) and its bound PABP play an important role in protecting mRNAs from ribonuclease attack (reviewed in Sachs, 1990; Ross, 1995). Furthermore, poly(A) facilitates translation (Sachs, 1990) and translation influences mRNA stability. Therefore, most polyadenylated mRNAs that are still engaged in ongoing translation are targets of the deadenylation-dependent pathway (Fig. 1). This pathway is initiated by shortening of the poly(A) tail, a process mediated by at least one poly(A) ribonuclease (PARN) in mammalian cells (Korner and Wahle, 1997; Korner et al., 1998).

The 5'-cap in mammalian cells, like that in yeast, also remains bound by translation initiation factor eIF4E in a stabilizing complex (Ptushkina et al., 1999). In the deadenylation-dependent pathways that have been characterized in yeast, poly(A) shortening is followed by decapping and both 5'-3' and 3'-5' exonucleolytic decay (reviewed in Beelman and Parker, 1995). In mammalian cells, however, only indirect evidence suggests that decapping may follow poly(A) shortening (Couttet et al., 1997). Decapping could permit 5'-3' degradation (Coutts and Brawerman, 1993) to occur perhaps simultaneously with 3'-5' degradation (Brewer, 1999). The isolation of a murine homologue of the yeast 5'-3' exoribonuclease, Xrn1p (Bashkirov et al., 1997), provides further support that mammalian decay pathways involve 5'-3' decay. However, for H4 histone mRNA, which lacks a poly(A) tail, 3'-5' degradation appears to be the major decay pathway (Caruccio and Ross, 1994).

The nonsense-mediated decay pathway (NMD) targets mRNAs with nonsense codons as part of a well-conserved physiological process involving mRNA surveillance. NMD prevents the accumulation of aberrant transcripts and truncated proteins by ensuring rapid decay of the mRNAs. Transcripts that harbor nonsense codons, unspliced introns or extended 3'-UTRs are all targets for the NMD pathway. In the yeast *S. cerevisiae*, this pathway involves deadenylation-independent decapping followed by 5'-3' degradation (reviewed in Culbertson, 1999; Hilleren and Parker, 1999). To our knowledge, the sequence of degradation steps involved in NMD in mammalian cells is unknown.

Some mammalian mRNAs like 9E3, insulin-like growth factor 2 (IGF2) and transferrin receptor (TfR) are degraded by endonucleolytic activities independent of deadenylation (Beelman and Parker, 1995). Following endonucleolytic cleavage, the resulting 5'-fragment is targeted for 3'-5' degradation activity, while the 3'-fragment may be a substrate for a 5'-3' decay activity (Fig. 1). A number of endoribonuclease activities and specific mRNA cleavage sites have long been under investigation (Ross, 1995). Only recently a sequence-specific endoribonuclease involved in the decay of α -globin mRNA has been identified in erythroid cells (Wang and Kiledjian, 2000a). The activity

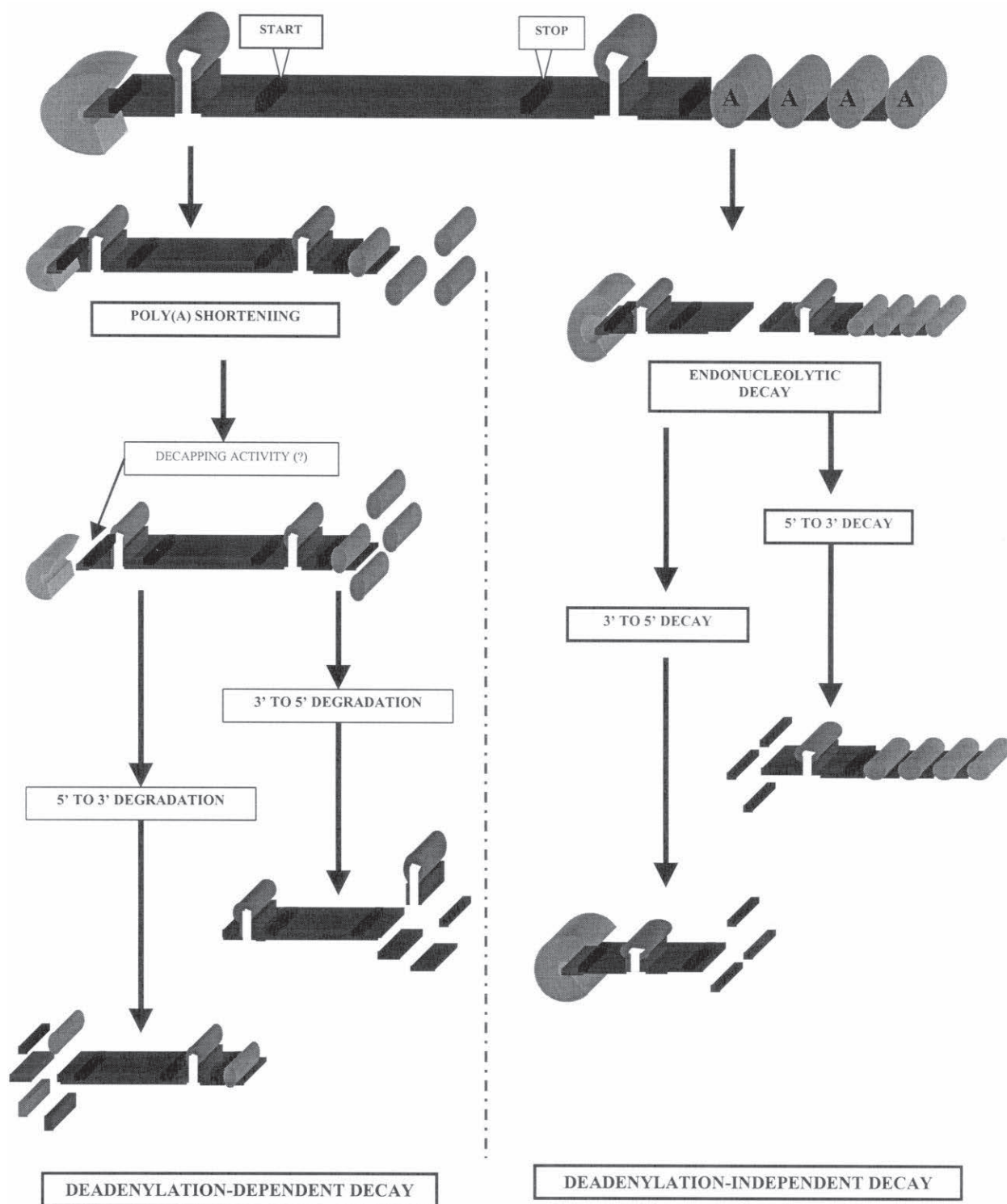


Fig. 1. Diagram of two mRNA decay pathways observed in mammalian cells. The mRNA molecule is shown with a 5' cap structure and stem loop structures in the 5' UTR and 3' UTR. The start and stop codons are indicated. The left panel shows deadenylation dependent decay, which appears to be the major pathway. Based on evidence presented by Couttet et al. (1997), decapping may occur after removal of most or all of the poly(A) tract. The identification of both 5' 3' and 3' 5' exoribonucleases suggests that the mRNA can be degraded bidirectionally. Examples include many ARE containing mRNAs. The right panel shows the deadenylation independent decay pathway. Endonucleolytic cleavage of an mRNA would generate two products that would likely be subject to exonucleolytic activities. Examples include transferrin receptor and insulin like growth factor 2 mRNAs.

is inhibited by the poly(A) tail specifically through an indirect interaction between PABP and an α -globin-specific stabilizing protein, α -CP (Wang and Kiledjian, 2000b). Some of the endonucleases are constitutively active and therefore their accessibility to the mRNA cleavage sites is under regulation. By contrast, others, like mammalian RNase L, which plays an important role in viral infection as well as mediating the interferon (IFN) response, are directly regulated (Li et al., 2000) (For further discussion, refer to Section 5.1).

4. Selected sequence determinants and their *trans*-acting proteins

Work during the past two decades has brought about an upsurge in the amount of information regarding the *cis*-acting structural determinants controlling mRNA turnover and the *trans*-acting stabilizing and destabilizing factors affecting regulation (Table 1) (Reviewed in Sachs, 1993; Ross, 1995; Liebhaber, 1997; Mitchell and Tollervey, 2000). In this section, we describe some selected, well-characterized RNA sequence elements that regulate mRNA decay to illustrate that these elements are not localized to just one region of mRNAs, but rather can be found throughout the body of mRNAs. Our discussion will also include the *trans*-acting factors that act through each element.

4.1. Elements in the 3'-UTR

The 3'-UTR has emerged as having a major role in harboring determinants that control mRNA decay. A few well-studied 3'-UTR determinants are listed below.

4.1.1. A + U-rich element (ARE) and proteins that bind AREs

The ARE has been recognized as a potent destabilizing element in a wide variety of short-lived mRNAs like those of proto-oncogenes and cytokines (reviewed in Chen and Shyu, 1995; Wilson and Brewer, 1999). The A + U-rich motif, located in the 3'-UTR of such mRNAs has been described by a variable number of often overlapping AUUUA pentamers, frequently harbored in or about a U-rich region. Although the features of a canonical ARE structure remain to be precisely defined, it is by now clear that inclusion of these sequences within 3'-UTRs of many mRNAs accelerates their decay. However, ARE-directed turnover mechanisms are heterogeneous in extent of decay as well as in decay characteristics. This heterogeneity presumably arises from the diverse RNA-binding factors recognizing the AREs and by the diverse RNA-protein and protein-protein interactions that presumably regulate turnover.

Several ARE-binding proteins have been described over the past decade (Wilson and Brewer, 1999). For brevity of this discussion, we limit our description to only a few well-characterized ones. The destabilizing potential of many

mRNAs is accelerated by a labile cytosolic activity (Loflin et al., 1999; Wilson and Brewer, 1999), first isolated from the erythroleukemic K562 cell line as AUF1 (Brewer, 1991). AUF1 is now known to exist as a family of four isoforms (p37, p40, p42 and p45) generated by alternative pre-mRNA splicing. AUF1 mediates ARE-directed decay in *trans* as part of a multi-subunit complex with itself (as homodimers or heterodimers of its isoforms) and with additional cellular factors, such as the translation initiation factor eIF4G, PABP, and the heat shock proteins hsp/hsc70 (Laroia et al., 1999). Although the destabilizing function of AUF1 is well-documented, recent studies suggest that AUF1 may also be involved in interactions with stabilizing complexes targeting specific sequence elements of a few mRNAs, such as the C-rich element (CRE) of α -globin mRNA (Kiledjian et al., 1997) and an A + U-rich sequence of the hypocalcemia-induced parathyroid hormone (PTH) mRNA (Sela-Brown et al., 2000). Besides AUF1, a class of mammalian proteins closely related to the *Drosophila* neuron-specific RNA-binding protein ELAV, including Hel-N1, HuC, HuD (Levine et al., 1993; Chung et al., 1996) and HuR (Ma et al., 1996) have been described that specifically bind to AREs and stabilize the mRNAs (Ma et al., 1997; Myer et al., 1997). Recently, tristetraproline (TTP), belonging to a family of zinc finger proteins that have two copies of the unusual CCCH zinc finger domain, has also been found to destabilize ARE-containing mRNAs such as TNF- α (Lai et al., 1999), GM-CSF (Carballo et al., 2000) and IL-3 mRNAs (Stoecklin et al., 2000). Another ARE-binding protein, first isolated from human spleen, specifically bound to the 3'-UTR of IFN- γ . It was found to be the glycolytic enzyme glyceraldehyde-3-phosphate dehydrogenase (GAPDH) (Nagy and Rigby, 1995). Other evidence for ARE-binding proteins with enzymatic functions has also been documented (Wilson and Brewer, 1999).

Besides the AREs that constitute a general class of instability determinants, a few other elements that have been characterized thus far appear mRNA-specific.

4.1.2. Iron-responsive element (IRE) and iron regulatory protein (IRP)

The iron-responsive element consists of a stem-loop structure with a 23–27 bp stem with a mismatched C and a 6-nucleotide loop with a C at its 5' end. The 3'-UTR of the transferrin receptor (TfR) mRNA contains five IREs, three of which are required to regulate its half-life, while the 5'-UTR of ferritin mRNA contains a single IRE that affects translation initiation. Two multifunctional, mRNA-specific, IRE-binding proteins are the iron regulatory proteins 1 and 2 (IRP1 and IRP2). When the intracellular iron concentration is low, the IRPs are upregulated by the assembly of an iron-sulfur cluster in IRP1 and by stabilization of the IRP2 protein. Thus, at low iron levels, the IRPs play two important roles, (1) they interact with the IREs in the 3'-UTR of TfR mRNA, thereby stabilizing it and allowing iron uptake, and, (2) they can bind to the IRE in the 5'-UTR of ferritin

Table 1

Selected examples of decay determinants and their interacting proteins regulating the turnover of a few mRNAs [See text for further details]

Location of determinant on mRNA	Decay determinant	Functional class/function of mRNA	mRNAs	Proteins that bind the determinant ^a	Ref
3'-UTR	A + U-rich elements	Immune regulators	Interleukins (such as, IL-1, IL-2, IL-3, IL-6) Interferons (such as, IFN- α)	AUF1 (D) ELAV proteins (S)	Carballo et al., 2000; Chen and Shyu, 1995; Lai et al., 1999; Loflin et al., 1999; Ma et al., 1997; Myer et al., 1997; Sirenko et al., 1997; Stocklin et al., 2000; Levy et al., 1996; Wilson and Brewer, 1999
	Proto-oncogenes		GM-CSF GRO- α TNF- α	TTP (D)	Chen and Shyu, 1995; Myer et al., 1997 Wilson and Brewer, 1999; Brewer, 1991; Brewer, 2000; Schiavi et al., 1992, 1994
	β -adrenergic receptor		β_1 -adrenergic receptor β_2 -adrenergic receptor	AUF1 (D) ELAV proteins (S)	Wilson and Brewer, 1999; Pende et al., 1996
	Regulated		VEGF in response to hypoxia PTH in response to calcium	HuR (S) AUF1 (S)	Levy et al., 1998 Sela-Brown et al., 2000
	Iron-responsive element	Maintenance of iron homeostasis	TfR	IRP (S in low iron and D in high iron)	Rouault and Klausner, 1997
Coding region	C-rich element	Cell-type specific	α -globin	α -CP (S) AUF1 (S) (component of the stabilizing complex)	Wilson and Brewer, 1999; Kiledjian et al., 1997
	Others	Miscellaneous	Histone stem-loop IGF II stem-loop	Histone-specific stem-loop binding protein Endonuclease cleavage site	Whitfield et al., 2000 Scheper et al., 1995
	JNK-response element	Immune regulators	c-myc c-fos β -tubulin	mRNA-specific proteins	Yeilding and Lee, 1997; Prokipcak et al., 1994 Schiavi et al., 1994; Chen et al., 1992 Bachurski et al., 1994
	5'-UTR		IL-2 Chemokine KC	Nucleolin (S) YB-1 (S) IL-1 α (S)	Chen et al., 2000 Tebo et al., 2000

^a Stabilizing, S; Destabilizing, D

mRNA, thereby imposing a block to translation initiation, preventing iron sequestration. On the other hand, when iron levels are high, the iron-sulfur cluster in IRP1 is disassembled and IRP2 is targeted for ubiquitin-directed proteolysis by the 26S-proteasome. This renders the TfR mRNA, now free of IRPs, more susceptible to endonucleolytic cleavage, thereby limiting iron uptake. At the same time, the translation initiation block on ferritin mRNA is also removed, thereby increasing levels of this iron sequestration protein (reviewed in Rouault and Klausner, 1997).

4.1.3. Cell cycle-regulated histone mRNA stem-loop determinant

Histone mRNAs lack poly(A) tails but their cell-cycle-dependent decay is mediated by a 3' terminal stem-loop motif (6 bp stem and a 4 base loop) (Pandey and Marzluff, 1987). A stem-loop binding protein, cloned from several species, and an uncharacterized labile factor are critical for the cell-cycle-dependence of histone mRNA regulation (Whitfield et al., 2000).

4.2. Elements in the coding region

Some mRNAs like c-fos, c-myc and β -tubulin contain decay elements in their coding regions. c-fos mRNA has two coding region determinants, CRD-1 and CRD-2, in addition to an ARE in its 3'-UTR (Chen et al., 1992; Schiavi et al., 1994). CRD-1, the major determinant, is a purine-rich 320 nt-segment that encodes a basic leucine zipper, critical for c-Fos protein function. However, it is the RNA sequence rather than the encoded polypeptide that is responsible for determining instability. A complex of proteins interacts with this determinant. These include the p37 isoform of AUF1; NSAP1, an hnRNP R-like protein; PABP; PAIP-1, which is a PABP-interacting protein; and Unr, a purine-rich RNA-binding protein (Grosset et al., 2000). However, the mechanism by which binding of this complex to CRD-1 directs mRNA degradation is not known.

Microtubules are dynamic filamentous structures that are composed of tubulin monomers. Tubulin is a heterodimer of α - and β - subunits that exist in dynamic equilibrium with the microtubule polymer. Tubulin synthesis is autoregulated by a post-transcriptional mechanism that renders tubulin mRNA levels sensitive to the intracellular concentration of tubulin subunits. β -tubulin mRNA stability, but not that of α -tubulin, is determined by the first 13 translated nucleotides, which encode the first four β -tubulin amino acids. Evidence suggests that the regulation of β -tubulin mRNA involves its cotranslational degradation (Bachurski et al., 1994). Moreover, the binding of one or more cellular factors might also be involved.

Previous studies have implicated the coding region determinant of c-myc mRNA, harbored within exon 3, and encoding the C-terminal 60 amino acids, to be responsible for the destabilization of c-myc mRNA during differentiation (Wisdom and Lee, 1991). Recent findings indicate that

coding sequences in both myc exon 2 and exon 3 are important in its downregulation during myoblast differentiation (Yeilding and Lee, 1997). A 70 kDa CRD-binding protein has been identified that specifically binds the c-myc exon 3 determinant and protects c-myc mRNA from endonucleolytic cleavage (Prokipcak et al., 1994).

4.3. Elements in the 5'-UTR

The 5'-UTR can play significant roles in altering stability either in a translation-dependent manner, by containing translation-inhibiting stem-loops, or in a manner independent of translation. An example of the latter is the stabilization of the otherwise unstable c-myc mRNA by reciprocal translocations of immunoglobulin introns in the 5'-UTRs of c-myc mRNA in certain lymphoma and plasmacytoma cells (Johnston and Carroll, 1992). Recent studies have also identified stability determinants within the 5'-UTR of the IL-2 mRNA (Chen et al., 2000) and the chemokine KC mRNA (Tebo et al., 2000). A JNK-response element (JRE) in the 5'-UTR of IL-2 mRNA is involved in its stabilization during T-cell activation. Two RNA-binding proteins, nucleolin and YB-1, specifically bind to the JRE and result in stabilization (Chen et al., 2000). In addition to a cluster of four overlapping AUUUA pentamers in the 3'-UTR, a 68-nt region in the 5'-UTR is required for the stabilization of KC mRNA (Tebo et al., 2000). Any trans-acting factor(s) mediating stabilization of KC mRNA are yet to be identified.

Two early observations, that most mRNAs are stabilized in the presence of translation inhibitors and that altering the 5'-UTR sequence, which affects translation initiation, also influences mRNA half-lives, have led to a thorough investigation for elucidating the possible link between mRNA stability and translation (Oliveira and McCarthy, 1995; Ross, 1995). There exists some evidence for an interaction between PABP and translation initiation factors in plant, yeast and animal cells (Gallie, 1998). Such an interaction is dependent on the phosphorylation state of PABP in certain plants (Le et al., 2000). Although the stabilization of some mRNAs in the presence of translation inhibitors is mediated through interactions with mRNA-specific regulatory factors, it has been proposed that a majority of mRNAs are stabilized by a common mechanism. It is believed that this mechanism represents a critical, evolutionarily conserved physiological process that occurs at each mitosis (Ross, 1997). The hypothesis envisages that translation repression during mitosis could inactivate mRNA decay pathways, stabilizing many labile mRNAs as well as their encoded proteins, thereby ensuring that these essential proteins are not depleted as the cell enters G1. Two additional, equally plausible explanations linking translation with mRNA decay are: (1) Ribosome translocation along mRNAs may deliver nucleases or expose nuclease sites; and (2) labile decay factors may be rapidly degraded following the cessation of translation.

5. Regulated mRNA stability

5.1. Regulation in response to development/differentiation cues

The c-myc proto-oncogene encodes a nuclear phosphoprotein that is a transcription factor regulating important genes involved in cell division and differentiation in a wide variety of cell lines (Henriksson and Luscher, 1996). c-myc expression is induced in proliferating cells (Kelly et al., 1983) but is downregulated during differentiation (Dony et al., 1985). The regulation of c-myc expression is mediated by post-transcriptional mechanisms in certain cell lines like C2C12 myoblasts (Wisdom and Lee, 1990), Daudi Burkitt's lymphoma cells (Dani et al., 1985), F9 teratoma cells (Dony et al., 1985) and K562 chronic myeloid leukemic cells (Brewer, 2000). Recent studies revealed that downregulation of c-myc mRNA induced during differentiation of C2C12 myoblasts to myotubes is mediated by a cytoplasmic mRNA turnover event, rather than a nuclear processing event (Yeilding and Lee, 1997). Both the coding region determinants of c-myc mRNA, located within exons 2 and 3, are required for c-myc downregulation (Yeilding and Lee, 1997). Another recent study on the destabilization of c-myc mRNA during the differentiation of K562 cells to megakaryoblasts, induced by TPA, implicates the involvement of polysome-associated, TPA-induced factors in mediating mRNA destabilization. This may also occur through interactions with the coding region determinants of c-myc mRNA (Brewer, 2000). Evidence also suggests that differentiation of K562 cells to megakaryoblasts may activate a deadenylation-independent pathway that targets the degradation of the body of c-myc mRNA.

MyoD belongs to a class of basic-helix-loop-helix transcription factors and is a key regulator of muscle cell differentiation (Olson, 1990). RNase L is a latent endoribonuclease implicated in the attenuation of IFN-stimulated responses including those induced by viral infection (Li et al., 1998). RNase L is activated by 2'-5' oligoadenylate (2-5A), generated in response to dsRNA (Zhou et al., 1993) and is inhibited by a specific protein called RNase L inhibitor (RLI) (Bisbal et al., 1995). Variations in activity of this pathway have been observed in growth and differentiation. A recent study demonstrates a role for RNase L and RLI in controlling the timing of muscle cell differentiation by orchestrating the regulation of MyoD mRNA stability (Bisbal et al., 2000). Use of chimeric constructs of RNase L and antisense RLI revealed that the effect of RNase L is to specifically lower MyoD mRNA levels and delay the differentiation of C2 myoblasts to myotubes. To our knowledge, this represents the first identification of a cellular mRNA target for RNase L.

5.2. Regulation in response to hormonal regulation

Steroid hormones control gene expression by binding and activating specific receptor proteins that then interact with

hormone response elements in the promoters of hormone responsive genes. The target mRNAs whose expression is controlled by regulating mRNA turnover in response to these hormones are numerous and are widely distributed. Interested readers are referred to an excellent review on the hormonal regulation of mRNA stability (Staton et al., 2000). The stabilization of various serum protein mRNAs, including vitellogenin and albumin mRNAs in *Xenopus*, is by far the best-studied examples of hormonal regulation of mRNA stability (Blume and Shapiro, 1989; Chernokalskaya et al., 1998). However, a considerable amount of information on this subject has also accumulated for mammalian cells.

Collagen is the most abundant protein in the body, and type I collagen constitutes the primary structural protein in bone. The relative stability of type I procollagen mRNA is greatly affected by glucocorticoid (Raghow et al., 1986) and growth factor treatment (Raghow et al., 1987). Procollagen mRNA is destabilized by dexamethasone and stabilized by transforming growth factor β (TGF β). The structural elements mediating transcriptional regulation of the type I procollagen gene have been characterized to some extent, but the knowledge of structural elements of the mRNA that are involved in the regulation of turnover is very limited. Glucocorticoids downregulate procollagen type I mRNA in fibroblasts through post-transcriptional mechanisms (Lindquist et al., 2000) as well as in osteoblasts both through transcriptional and post-transcriptional mechanisms. The post-transcriptional regulation of procollagen type I mRNA by glucocorticoids is mediated by the regulation of mRNA turnover. However, cortisol decreases the stability of α 1 procollagen in transcription-arrested osteoblast cells (Delany et al., 1995). The 3'-UTR contains highly conserved regions that presumably play a role in this regulation. Recently, a 67 kDa, cytoplasmic, RNA-binding activity, α 1-RBF₆₇ was identified in NIH3T3 cells and in human skin fibroblasts. This activity specifically binds to the 3'-UTR conserved region (Maatta and Penttinen, 1994; Mahonen et al., 1998) and is therefore implicated in the post-transcriptional regulation of procollagen type I mRNA turnover. This study has important pathophysiological implications in that excess glucocorticoids induce the development of osteoporosis by two mechanisms, (1) by inhibiting cell replication, which depletes the population of cells synthesizing type I collagen, and (2) by downregulating collagen gene expression in osteoblasts.

T₃ regulates TSH- β expression by both transcriptional (Shupnik et al., 1985) as well as post-transcriptional mechanisms involving mRNA turnover (Staton and Leedman, 1998). A 70 nucleotide (nt), cis-acting element was recently identified in the 3'-UTR of the TSH- β mRNA (Staton et al., 2000). Although it does not contain ARE-like sequences or any other known cis-acting decay elements (Leedman et al., 1995), the TSH- β specific determinant functions through the interaction of T₃-regulated RNA-binding proteins present in murine thyrotrope cells

(TSH- β binding protein 1 and 2) (Staton et al., 2000). Additionally, PABP may interact with the TSH- β binding proteins (TSH- β BPs). It has therefore been proposed that T3 regulates the binding of TSH- β BPs and PABP to the 3'-UTR of the TSH- β mRNA, thereby stimulating mRNA deadenylation and decay. However, in the absence of T₃, the mRNA remains polyadenylated and stable.

Glucocorticoids play an important role in the treatment of leukemias and other malignant and non-malignant lymphoproliferative diseases (Harmon et al., 1979). They are also implicated in triggering apoptosis in CD4 + /CD8 + cells in the thymus (Cohen, 1992). Glucocorticoids have been proposed to induce G₀/G₁ arrest by downregulating the expression of c-myc and cyclin D₃, two components critical for G₁ progression (Reisman and Thompson, 1995). The control of cyclin D₃ expression by glucocorticoids is achieved by decreasing mRNA stability. A recent study of cyclin D₃ regulation by glucocorticoids identified an element, called the BC element, located in the center of two polypyrimidine tracts in the 3'-UTR of the cyclin D₃ mRNA. This element is highly conserved in mouse and human and assumes a stable stem-loop structure (Garcia-Gras et al., 2000). Two proteins of approximately 37 and 52 kDa that bind to the BC element may be involved in cyclin D₃ regulation in P1798 murine T lymphoma cells. Two observations, that cyclin D₃ mRNA lacking a 3'-UTR is not destabilized by glucocorticoids and that with overexpression of the BC element, cyclin D₃ mRNA loses sensitivity to glucocorticoids, have led to the hypothesis that formation of a ternary complex involving the BC element and the two proteins is required, but is not sufficient, for the destabilization of cyclin D₃ mRNA. It has been proposed further that the ternary complex recruits an additional, unidentified RNA-destabilizing or RNA-degrading activity, forming an unstable quaternary complex that is eventually responsible for degradation.

5.3. Regulation in response to diurnal variation

The cholesterol 7- α -hydroxylase (CYP7A1) mRNA, protein and enzyme activity display rapid changes in response to diurnal variation (Noshiro et al., 1990; Lavery and Schibler, 1993; Lee et al., 1994). The enzyme catalyzes the initial step in the conversion of cholesterol to bile acid. Earlier studies indicated that alterations in CYP7A1 gene transcription might regulate its expression. However, a recent study demonstrated a >20-fold increase in steady-state levels of CYP7A1 mRNA in L35 hepatoma cells following dexamethasone treatment, suggesting the involvement of mRNA stabilization in this regulatory scenario (Trawick et al., 1997). Moreover, multiple AUUUA motifs, characteristic of many AREs, are conserved in the 3'-UTR of the CYP7A1 mRNA from several species. Therefore, the 3'-UTR of CYP7A1 mRNA may mediate rapid turnover in response to diurnal changes. Further studies identified a labile protein that is depleted from transcriptionally blocked

cells, which is sufficient to mediate dexamethasone-induced stabilization (Baker et al., 2000). This regulation may prevent the accumulation of cytotoxic bile acids and control the cellular pool of cholesterol in hepatocytes, which is intimately linked to CYP7A1 expression. Thus, the concordant regulation of CYP7A1 transcription and mRNA degradation assists the cell in adapting rapidly to the complex metabolic demands of the liver.

5.4. Regulation in response to stress

Lactate dehydrogenase (LDH) is a key enzyme in anaerobic glycolysis. Stress-induced activation of β -adrenergic receptors by hypoxia (Semenza et al., 1996), phorbol ester (Huang and Jungmann, 1995) or c-myc-induced tumorigenesis (Shim et al., 1997) causes a shift in LDH isozyme pattern in favor of the homotetrameric (A₄) LDH-A isozyme 5. The stress-induced upregulation of LDH-A involves both transcriptional and post-transcriptional mechanisms, primarily mRNA stabilization, that are mediated synergistically by protein kinase A (PKA) and protein kinase C (PKC) (Huang and Jungmann, 1995). A recent study identified a cAMP-stabilizing region (CSR) within the 3'-UTR of the LDH mRNA. When the CSR interacts with specific CSR-binding proteins, LDH-A mRNA is stabilized. This occurs in response to PKA activation through phosphorylation/dephosphorylation cascades mediated by PKA and an unidentified phosphatase (Short et al., 2000). Mutation analyses and ribonuclease protection assays identified a uridine-rich region within the LDH-A 3'-UTR, termed the PKC-stabilizing region (PCSR), which mediates stabilization in response to PKC. Interestingly, the CSR and PCSR overlap in a common 13 nt stretch. Based on these observations, it was suggested that a synergistic interaction of PKA and PKC stabilizes LDH-A mRNA. This is biologically significant because the LDH-A isozyme 5, with a low K_m for pyruvate and a high V_{max}, can derive more energy from an anaerobic cell by reducing pyruvate to lactate and concomitantly producing NAD⁺, thereby fulfilling the requirement for increased glycolysis by tumor and hypoxic cells.

Hypoxia also induces vascular endothelial growth factor (VEGF) through mRNA stabilization (Levy et al., 1996), in addition to transcriptional (Mukhopadhyay et al., 1997) and translational activation (Stein et al., 1998). VEGF is implicated in normal as well as pathological angiogenic processes (Risau, 1997). A recent study demonstrated that hypoxia-induced stabilization of VEGF mRNA is mediated through the activation of stress-activated protein kinases like p38 and JNK (Pages et al., 2000). The regulation is generally achieved through the interaction of different proteins with various parts of the VEGF 3'-UTR. An ARE with consensus domains UUAUUUA(U/A)(U/A) is present in the 3'-UTR, one of which may form a stable hairpin loop. It has been suggested that JNK exerts its effects through this loop by recruiting various proteins. One such protein is the RNA-binding protein HuR, which binds to the VEGF ARE with

high affinity and specificity and mediates hypoxia-induced stabilization (Levy et al., 1998).

Parathyroid hormone (PTH) regulates calcium and phosphate homeostasis and is in turn controlled by serum calcium and serum phosphate levels (Moallem et al., 1998). Hypocalcemia increases both PTH secretion and mRNA levels and eventually PT cell proliferation. Calcium levels are restored by PTH-mediated mobilization of calcium from bone and renal reabsorption. On the other hand, hypophosphatemia decreases serum PTH, PTH mRNA levels, and parathyroid (PT) cell proliferation. PTH readjusts phosphate levels by increasing renal phosphate excretion. Understanding this regulation is important, since patients with chronic renal failure develop severe complications associated with hyperparathyroidism, which is stimulated by high levels of serum phosphate in these patients (Sela-Brown et al., 2000). Both these inverse regulatory effects are mediated by post-transcriptional mechanisms, specifically involving the same mRNA decay pathway. Previous studies suggested the involvement of PT cellular proteins and their interaction with the 3'-UTR of PTH mRNA to mediate calcium or phosphate induced regulation (Moallem et al., 1998). Based on these studies, it was suggested that hypocalcemic-induced proteins stabilize the PTH transcript, which is destabilized by hypophosphatemic-induced proteins. RNA affinity chromatography and microsequencing identified one of the PT cellular proteins that binds to the PTH 3'-UTR as AUF1. In vitro studies suggested a stabilizing function of AUF1 for the PTH transcript, involving an A + U-rich sequence in the 3'-UTR (Sela-Brown et al., 2000). It was proposed that hypocalcemia stabilizes PTH mRNA by increasing the binding of AUF1 and other PT cellular proteins to the 3'-UTR A + U-rich region. By contrast, hypophosphatemia destabilizes the PTH mRNA by decreasing this specific binding activity.

6. Deregulated mRNA stability

6.1. Deregulation in carcinoma

Cyclooxygenase catalyzes the rate-limiting step in the conversion of free arachidonic acid to prostaglandins that have been implicated in normal growth responses as well as carcinogenesis (Marnett, 1992; Prescott and White, 1996). There are two isoforms of cyclooxygenase. Cyclooxygenase 1 (COX1) is constitutively active and makes prostaglandins for normal physiological function. Cyclooxygenase 2 (COX2), not normally present under basal conditions, is rapidly induced by growth factors, cytokines and tumor promoters, which results in aberrant prostaglandin synthesis associated with inflammation and carcinogenesis. The presence of multiple copies of AUUUA pentamers in the 3'-UTR of COX2 mRNA suggested the involvement of mRNA stabilization in the tumor-induced upregulation of

COX2 (Dixon et al., 2000). Further studies have demonstrated that the upregulation is achieved through the interaction of ARE-binding proteins with a conserved 116 nt ARE (Lasa et al., 2000). A recent study has also implicated the induction of the mitogen activated protein kinase p38 signaling cascade to be responsible for stabilizing COX2 mRNA in tumor cells (Lasa et al., 2000).

Genetic alterations in ARE and ARE-binding regulatory proteins presumably play a role in neoplastic transformation. Identification of such mutations and studying how they alter regulation in cancer cells would shed light on the intriguing mechanisms involved in tumorigenicity and help in developing novel modes of therapy. One such mode of chemotherapy employing prostaglandin A₂ (PGA₂) arrests growth at the G1 phase of the cell cycle in many cancer cells. PGA₂ achieves this growth arrest by down-regulating cyclin D1 expression, which is crucial for G1 progression. A recent study investigating the mechanisms that lead to this downregulation suggests that AUF1 and other RNA-binding proteins may be involved in interactions with a 390-nt sequence in the 3'-UTR of cyclin D1 mRNA. Such destabilizing interactions could be responsible for cyclin D1 downregulation by PGA₂ (Lin et al., 2000).

6.2. Deregulation in Alzheimer's disease

In Alzheimer's disease, iron is thought to accumulate in the brain without a concomitant increase in ferritin (Pinero et al., 2000). Such an aberrant accumulation of iron without proper sequestration can render cells vulnerable to oxidative stress, an important component of Alzheimer's disease. As mentioned earlier, iron homeostasis in the cell is achieved by concomitant regulation of ferritin, the iron sequestration protein, and transferrin receptor (TfR), the iron uptake protein, allowing the cell to obtain iron when needed and sequester iron when in excess (Section 4.1.2.). It has been hypothesized that alterations in the IRE-IRP interaction could be responsible for the aberrant iron accumulation in Alzheimer's brains. A recent study addressing this question demonstrated the presence of a single IRE/IRP complex with decreased mobility and increased stability in brain extracts of patients with Alzheimer's disease, compared to a double complex in normal brain extracts, as observed by RNA band shifts and other in vitro assays. It has therefore been suggested that a more stable IRE/IRP complex in Alzheimer's disease brains could stabilize TfR mRNA while inhibiting ferritin synthesis. Such an alteration would then result in increased iron uptake without concomitant iron sequestration by ferritin, as is observed in brains of patients with Alzheimer's disease. A mutation in either the IRE or IRP or both could be responsible for increasing this binding. An inhibitory mutation involving the ubiquitin-directed proteolytic pathway targeting IRP in high iron could also be involved.

7. Conclusion and perspectives

The regulation of mRNA stability is one of nature's intriguing and very effective molecular adaptations that enables the cell to maintain a translatable transcript without embarking upon the energy-demanding steps of transcription, processing and export. Additionally, it permits a cell to respond rapidly to changes in intrinsic and extrinsic stimuli. As we have seen, mammalian cells modulate the rate of mRNA turnover in order to regulate a wide variety of transcripts that have important physiological and pathophysiological functions. We know something about mammalian mRNA decay pathways and some of the enzymatic activities responsible. We also know that cis-acting sequence elements in mRNAs exert their actions through interactions with trans-acting factors. However, many details remain to be understood. For example, exactly how does binding of a trans-acting factor to a cis element promote mRNA degradation or stabilization? Presumably these factors must provide signals to the mRNA degradation machinery, but the question is how. This will necessitate the identification of all the enzymatic components and cofactors of this machinery. Moreover, many of the trans-acting proteins identified so far seem to shuttle between the nucleus and cytoplasm. What role does this process play in mRNA decay? Does higher order RNA structure within cis-acting elements affect interactions with trans-acting factors? What are the signal transduction pathways that affect mRNA decay and which components of the decay machinery are the direct targets of these pathways? Clearly, there is still much to be learned. Perhaps this knowledge will aid in the design of novel approaches to therapeutic intervention in the various abnormalities that are associated with deregulated mRNA stability.

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